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1. Convex lens

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2. Power is inversely proportional to focal length. So a convex lens of focal length 10 cm has a greater power.

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3. Given :

Object is placed 5 cm in front of the lens.

The image is formed behind the lens at 10 cm.

$$\therefore u = 5 \text{ cm}$$

$$v = 10 \text{ cm}$$

Since the image formed behind the mirror and the image formed at a distance double the object placed, the lens is a convex lens.

Now ,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{10} - \frac{1}{-5}$$

$$\frac{1}{f} = \frac{1+2}{10}$$

$$\frac{1}{f} = \frac{3}{10}$$

$$f = \frac{10}{3}$$

$\therefore$ The value of its focal length is $\frac{10}{3}$ cm.

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4. Given :

$$v = 50 \text{ cm}$$

$$h_i = h_o$$

$$m = -1$$

$$m = \frac{v}{u}$$

$$-1 = \frac{50}{u}$$

$$u = 50 \text{ cm}$$

Lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{50} - \frac{1}{-50}$$

$$\frac{1}{f} = \frac{2}{50}$$

$$\frac{1}{f} = \frac{1}{25}$$

$$f = 25 \text{ cm}$$

Now,

$$P = \frac{1}{f_m} = \frac{1}{25 \times 10^{-2}}$$

$$1\text{D} = 1 \text{ m}^{-1}$$

$$4\text{m}^{-1} = 4\text{D}$$

$\therefore$ The object is placed at a distance of 50 cm from the lens and the power of the lens is 4D.

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